

TE 454 · Digital Signal Processing

Compact Quick Reference — Lectures 1-3 (Introduction, Continuous-Time, Discrete-Time)

Lecture 1 · Introduction to Digital Signal Processing

1.1 What is DSP?

Digital Signal Processing: theories, methods and algorithms for processing signals in digital form. **Digital Signal Processor:** a microprocessor built to run DSP algorithms (architecture: chips, peripherals, pipelines, instructions; plus hardware & software system design).

Analog signal → continuous in both time and magnitude. **Digital** signal → discrete in both time and magnitude. Most real-world signals are analog, so DSP needs a conversion: **Step 1 – Sample** the analog signal at discrete time points; **Step 2 – Quantize** the sample values (by rounding or truncation).

Why DSP matters: digitization is the foundation of information technology, and DSP is the kernel of digitization. Most real-time DSP is implemented on DSP processors or DSP-core ASICs — a key driver of the whole semiconductor industry.

1.2 Digital vs Analog Signal Processing

Advantages of digital over analog: more reliable / less sensitive to component tolerances (temperature, noise, ageing); higher accuracy; can be integrated on a single chip. **Limitation:** A/D and D/A conversion rates are not available at every extreme — beyond ~GHz, or below a few Hz, converters become impractical.

· Why Digital? — Explanation

- 1. Programmability.** Analog systems need a hardware redesign to change behaviour; digital systems only need new software (e.g. adaptive filters).
- 2. Precision.** Analog precision is set by component spec (resistors $\pm 5\%$, capacitors $\pm 20\%$ or worse); digital precision is set by ADC bit-depth, word length and the algorithm.
- 3. Stability.** Analog components (R, C, op-amps) drift with temperature/humidity; digital systems show no variation over their guaranteed operating range.
- 4. Anti-noise.** Digital signals tolerate noise far better since only two logic levels must be distinguished.
- 5. Repeatability.** Two digital implementations of the same design behave identically; two analog builds (component variation) do not.
- 6. VLSI.** Digital algorithms map efficiently onto very-large-scale integrated circuits.
- 7. Error-correcting codes.** Digital/binary data can carry redundant bits so transmission/storage errors are detected and corrected.
- 8. Data transmission & storage.** Digital media (Internet, CD, DVD) give far higher fidelity than analog storage/transmission.
- 9. Data compression.** Digital techniques allow lossless compression; analog compression (e.g. 3 kHz-limited telephone lines) always loses information.

1.3 We Still Need Analog Processing

- **Real-time processing:** analog processing time is only the circuit delay; digital processing time depends on processor speed and algorithm — harder to guarantee true real time.
- **Very high frequency signals:** analog circuits can process microwave/mm-wave/light-wave signals directly; by the Nyquist rule, digital processing is capped by sample-and-hold and A/D speed, and very-high-rate ADCs are expensive.
- **Most real-world signals are analog:** to process them digitally they must first be converted via mixed-signal (analog + digital) processing.

1.4 Applications of DSP

Area — Examples

Communication. Wireless comms (base stations, switching centres, mobile phones); GSM cell-phone chain

Satellite navigation. GPS (Global Positioning System), GIS (Geographic Information System)

Measurement & control. Virtual instruments (e.g. vector analyzers), automotive electronics

Military & safety. Early-warning radar/sonar, cruise missiles, smart bombs, UAVs

Consumer electronics. Digital video/camera, MP3, HDTV, home theatre, IP phone, IPTV

Medical. Ultrasound, MRI, CT, gamma knife, hearing aids

Other. Drowsy-driving alert, digital watermarking, video surveillance, fingerprint ID, pattern recognition, image/video repair

Course context: assessment = Exams 50% + Project 20% + Homework 20% + Quizzes 10%; related courses are Signals & Systems, Advanced Maths/Linear Algebra, Programming, Microcomputer Principles, Logic Circuits.

Lecture 2 · Continuous-Time Signals and Systems

2.1 Analysis Philosophy

There are too many distinct signals/systems to study one at a time. Instead: decompose an arbitrary signal into simple building-block signals, find the system's response to each, then combine (superpose) those responses to get the response to the original input. This is the core idea behind both time-domain (impulse/convolution) and frequency-domain (Fourier) analysis.

2.2 Typical Continuous-Time Signals

A signal is a time function $f(t)$. If $f(t) = 0$ for $t < 0$ it is a **causal** signal (has a start at $t = 0$).

Signal — Definition / notes

Unit step $u(t)$. $u(t)=1$ for $t \geq 0$, 0 otherwise. Basic causal signal — multiplying any signal by $u(t)$ makes it causal.

Sinusoid. $f(t)=A \sin(\omega t+\phi)=A \sin(2\pi f t+\phi)$. A =amplitude, f =frequency (Hz), $\omega=2\pi f$ =angular frequency (rad/s), ϕ =start phase. Non-causal & periodic: $f(t)=f(t+mT)$; representable via Euler's relation (complex exponentials).

Exponential, real α : $f(t)=e^{\alpha t}$. $\alpha < 0$ decaying, $\alpha = 0$ constant, $\alpha > 0$ growing.

Exponential, complex $\alpha=\sigma+j\omega$. $f(t)=Ae^{\alpha t}=Ae^{\sigma t}\cos\omega t + jAe^{\sigma t}\sin\omega t$. $\sigma=0$: sinusoidal; $\sigma>0$: growing sinusoid; $\sigma<0$: decaying (damped) sinusoid.

Gate $pr(t)$. $pr(t)=u(t+\tau/2) - u(t-\tau/2)$: a rectangular pulse of width τ , built from two shifted unit steps.

2.3 Unit Impulse $\delta(t)$ and its Properties

$\delta(t)$ is zero everywhere except $t=0$, and cannot be given a finite value even at $t=0$ — it is a **singularity (generalized) function**, defined only through an integral, not a regular function with exact point values. It is obtained as the limit of a gate pulse $pr(t)$ whose width $\tau \rightarrow 0$ while its area stays 1 (so its height $\rightarrow \infty$).

Property — Statement

Sampling property. $f(t) \cdot \delta(t) = f(0)\delta(t)$ (since $\delta(t)=0$ for $t \neq 0$ and $f(t)$ is constant at $t=0$); $\int f(t)\delta(t)dt = f(0)$

Shift. $\delta(t-\tau)$ is the impulse relocated to $t=\tau$

Scaling. $A\delta(t)$: A is the impulse *intensity* = area under the impulse

Derivative / integral. $\delta'(t)=d\delta(t)/dt$ is also an impulse-type function; the integral of $\delta(t)$ is the unit step $u(t)$

Even symmetry. $\delta(t)=\delta(-t)$, inherited from the even gate signal it derives from

Any signal can be approximated as a sum of narrow rectangles of width $\Delta\tau$, height $f(k\Delta\tau)$ — i.e. weighted, shifted impulses $f(k\Delta\tau)\Delta\tau\delta(t-k\Delta\tau)$. As $\Delta\tau \rightarrow d\tau$ this sum becomes the exact integral representation of $f(t)$ as a continuum of shifted, weighted impulses — the basis for convolution.

2.4 Linear Time-Invariant (LTI) Systems & Convolution

Property — Condition

Linear. Scaling: $af(t) \rightarrow ay(t)$. Additivity: $f_1(t)+f_2(t) \rightarrow y_1(t)+y_2(t)$. Combined: $af_1(t)+bf_2(t) \rightarrow ay_1(t)+by_2(t)$

Time-invariant. $f(t-t_0) \rightarrow y(t-t_0)$ for any shift t_0

The response to $\delta(t)$ is the system's **unit impulse response $h(t)$** . Because any input $f(t)$ is a sum of shifted, weighted impulses, and the system is linear & time-invariant, the total response is the same sum of shifted, weighted copies of $h(t)$ — this sum is the **convolution** of $f(t)$ and $h(t)$:

$$y(t) = f(t) * h(t) \text{ (convolution)}$$

Time-domain view: decompose signals into shifted/weighted $\delta(t)$ and track only the impulse response. Frequency-domain view: decompose signals into $\sin\omega t$ and its harmonics instead (Fourier analysis, below).

2.5 Fourier Series (Periodic Signals)

A periodic signal (period T) can be expanded as a Fourier series (trigonometric or, more compactly, exponential form), with coefficients equal to the signal's average over one period. Example: a periodic rectangular pulse train (period T , pulse width τ , amplitude A) has Fourier coefficients

$$F_n = (A\tau/T) \cdot \text{Sa}(n\omega_0\tau/2), \text{Sa}(x) = \sin(x)/x, \omega_0=2\pi/T$$

where $\text{Sa}(x)=\sin(x)/x$ is the **sampling function** — the discrete line spectrum of a rectangular pulse train follows its envelope.

2.6 Fourier Transform (Non-Periodic Signals)

Viewing a non-periodic signal as the limit of a periodic one as $T \rightarrow \infty$: the harmonic spacing $\omega_0 \rightarrow 0$ and each line's amplitude $F_n \rightarrow 0$, so discrete spectral lines no longer make sense — instead we describe the signal by a continuous **spectrum density**, the Fourier Transform $F(j\omega)$.

Signal — Fourier Transform $F(j\omega)$

Gate $pr(t)$. $\tau \cdot \text{Sa}(\omega\tau/2)$ — a Sa-shaped spectrum, first zeros at $\omega=\pm 2\pi/\tau$

Exponential $e^{-at}u(t)$, $a>0$. $1/(a+j\omega)$; magnitude peaks at $1/a$, decays with ω

Unit impulse $\delta(t)$.	1 — uniform (flat) density over all frequencies $\Rightarrow$ infinite bandwidth
Constant 1 (DC).	$2\pi\delta(\omega)$ — all energy concentrated at $\omega=0$ (symmetry of the DC/impulse pair)
$\cos\omega_0 t$.	$\pi[\delta(\omega+\omega_0)+\delta(\omega-\omega_0)]$ — two real impulses at $\pm\omega_0$
$\sin\omega_0 t$.	$j\pi[\delta(\omega+\omega_0)-\delta(\omega-\omega_0)]$ — imaginary impulses at $\pm\omega_0$
Unit impulse train (period T).	another impulse train in frequency, spacing $\omega_0=2\pi/T$, weight ω_0 — basis of sampling theory

2.7 Properties of the Fourier Transform

Property — Statement	
Linearity.	$af_1(t)+bf_2(t) \leftrightarrow aF_1(j\omega)+bF_2(j\omega)$ (FT is an integral $\Rightarrow$ linear operator)
Time shift.	$f(t-t_0) \leftrightarrow F(j\omega)e^{-j\omega t_0}$ — a time shift is a pure phase shift in frequency
Frequency shift (modulation).	$f(t)e^{j\omega_0 t} \leftrightarrow F[j(\omega-\omega_0)]$; so $f(t)\cos\omega_0 t \leftrightarrow \frac{1}{2}\{F[j(\omega+\omega_0)]+F[j(\omega-\omega_0)]\}$ — amplitude modulation shifts the spectrum to $\pm\omega_0$
Energy / Parseval.	Signal energy W computed in time equals the integral of $ F(j\omega) ^2$ (energy spectrum) in frequency — same shape as $ F(j\omega) $ but with no phase information
Convolution (time).	$f_1(t)f_2(t) \leftrightarrow F_1(j\omega)F_2(j\omega)$ — convolution in time = multiplication in frequency
Convolution (frequency).	$f_1(t)f_2(t) \leftrightarrow (1/2\pi)[F_1(j\omega)*F_2(j\omega)]$ — multiplication in time = convolution in frequency

The convolution theorem is the key link: it turns convolution in one domain into plain algebra (multiplication) in the other — nearly every other FT property can be derived from it.

2.8 Frequency-Domain Analysis of Linear Systems

$H(j\omega)$ is the system (transfer) function: $Y(j\omega)=F(j\omega)H(j\omega)$, the frequency-domain counterpart of $y(t)=f(t)*h(t)$. When a system is too complex for an analytical $H(j\omega)$, it can be measured empirically by driving the system with sinusoids of different frequencies and recording the output — this technique is called a **frequency sweep**.

Lecture 3 · Discrete-Time Signals and Systems

3.1 Discrete-Time Signals & Sampling

A discrete-time sequence $x[n]$ is often generated by uniformly sampling a continuous-time signal $x_a(t)$:

$$x[n] = x_a(t)|_{t=nT} = x_a(nT), \quad n = \dots, -1, 0, 1, \dots$$

T = sampling interval/period; $F_T = 1/T$ = sampling frequency. **Sampled-data signal**: samples are continuous-valued. **Digital signal**: samples are discrete-valued (obtained by quantizing via rounding/truncation) — real DSP systems work on digital signals. A sequence with zero samples for $n < N_1$ is **right-sided**; if $N_1 \geq 0$ it is **causal**.

3.2 Basic Operations on Sequences

Operation — Definition

Product (modulation). $y[n] = x[n] \cdot w[n]$

Addition. $y[n] = x[n] + w[n]$

Scalar multiplication. $y[n] = A \cdot x[n]$

Time shift — unit delay. $y[n] = x[n-1]$

Time shift — unit advance. $y[n] = x[n+1]$

Time reversal (folding). $y[n] = x[-n]$

Branching. one sequence copied to feed multiple operations

Example combining several: $y[n] = \alpha_1 x[n] + \alpha_2 x[n-1] + \alpha_3 x[n-2] + \alpha_4 x[n-3]$.

3.3 Sampling-Rate Alteration

Generates a new sequence $y[n]$ with sampling rate F_T different from $x[n]$'s rate F_T . Alteration ratio $R = F_T'/F_T$: **$R > 1 \Rightarrow$ interpolation** (rate increases), **$R < 1 \Rightarrow$ decimation** (rate decreases).

3.4 Classifying Sequences

Class — Definition

Periodic. $x[n] = x[n+N]$ for all n — satisfies the periodicity condition for some integer N

Energy signal. Finite total energy $\sum |x[n]|^2$, but zero average power (e.g. any finite-length sequence)

Power signal. Infinite total energy but finite average power (e.g. a periodic sequence)

Bounded. $|x[n]| \leq B_x < \infty$ for all n (e.g. $\cos(0.3\pi n)$ is bounded)

Absolutely summable. $\sum |x[n]| < \infty$ over all n

Square summable. $\sum |x[n]|^2 < \infty$ (finite energy) over all n

3.5 Basic Sequences

Sequence — Definition / notes

Unit sample $\delta[n]$. 1 at $n=0$, 0 elsewhere — discrete analogue of $\delta(t)$

Unit step $u[n]$. 1 for $n \geq 0$, 0 for $n < 0$

Real sinusoid. $x[n] = A \cos(\omega_0 n + \phi)$: A =amplitude, ω_0 =angular frequency, ϕ =phase

Exponential. $x[n] = A\alpha^n$, A and α real or complex. Real α : $|\alpha| > 1$ grows (e.g. $\alpha = 1.2$), $|\alpha| < 1$ decays (e.g. $\alpha = 0.9$). Complex: real/imag parts are sinusoids with constant ($\sigma_0 = 0$), growing ($\sigma_0 > 0$) or decaying ($\sigma_0 < 0$) envelopes

$A \cos(\omega_0 n + \phi)$ and $B e^{j\omega_0 n}$ are periodic with period N iff $\omega_0 N = 2\pi r$ for positive integers N, r ; the smallest such N is the **fundamental period** (e.g. $\omega_0 = 0.1\pi \Rightarrow N = 2\pi r / \omega_0 = 20$ for $r=1$). As with continuous signals, an arbitrary sequence can be written as a weighted sum of basic sequences and their delayed/advanced versions.

3.6 The Sampling Process & Nyquist Theorem

Sample instants: $t_n = nT = n/F_T = 2\pi n/\Omega_T$, where $\Omega_T = 2\pi F_T$ is the sampling angular frequency. For a signal $x_a(t)$ built from a weighted sum of sinusoids, $\{x[n]\}$ represents $x_a(t)$ uniquely only if Ω_T is chosen greater than $2 \times$ the highest frequency present in $x_a(t)$. This anti-aliasing condition is the **sampling theorem (Nyquist theorem)**.

3.7 Discrete-Time System Properties

Property — Definition

Linear. Input $ax_1[n] + bx_2[n] \Rightarrow$ output $ay_1[n] + by_2[n]$, for any constants a, b and any inputs

Shift-invariant. $x[n] = x_1[n-n_0] \Rightarrow y[n] = y_1[n-n_0]$ for any integer n_0 (a.k.a. time-invariant)

Causal. Output at n depends only on present and past inputs

Stable. Bounded input produces bounded output

Passive / lossless. Output energy does not exceed (passive) / exactly equals (lossless) input energy

A system satisfying both linearity and shift-invariance is an **LTI system** — mathematically easy to analyze/design, and the basis of most practical DSP algorithms.

3.8 Impulse Response & the Convolution Sum

The response to $\delta[n]$ is the **impulse response $h[n]$** ; the response to $u[n]$ is the **step response $s[n]$** . Because linearity+shift-invariance let any $x[n]$ be written as a sum of weighted, shifted unit samples, the output is fully determined by $h[n]$:

$y[n] = x[n] * h[n] = \sum_k x[k]h[n-k]$ (convolution sum)

Property — Statement
Commutative. $x[n]*h[n] = h[n]*x[n]$
Associative. $(x[n]*h[n])*y[n] = x[n]*(h[n]*y[n])$
Distributive. $x[n]*(h[n]+y[n]) = x[n]*h[n] + x[n]*y[n]$

Graphical procedure to evaluate $y[n]$: (1) time-reverse $h[k]$ to get $h[-k]$; (2) shift right by n ($n>0$) or left ($n<0$) to get $h[n-k]$; (3) form the product $v[k]=x[k]h[n-k]$; (4) sum all samples of $v[k]$ to get $y[n]$. If $x[n]$ and $h[n]$ have lengths M and N , the convolution result has length **$M+N-1$** .

3.9 FIR vs IIR, Recursive vs Non-Recursive

Classification — Definition
FIR (Finite Impulse Response). $h[n]=0$ for $n<N_1$ and $n>N_2$ (finite length). Output is a finite sum of products, computed directly from the convolution sum.
IIR (Infinite Impulse Response). $h[n]$ has infinite length; characterized by linear constant-coefficient difference equations. Example — the accumulator $y[n]=y[n-1]+x[n]$ is IIR.
Non-recursive. Output computed from present & past <i>inputs</i> only
Recursive. Output computation also uses past <i>output</i> samples (feedback), in addition to present/past inputs